CODECHECK certificate

Reviewed paper: Harisena, N. V., Groen, T. A., Toxopeus, A. G., & Naimi, B. (2021). When is variable importance estimation in species distribution modelling affected by spatial correlation? *Ecography*, 44(5), 778-788. DOI: https://doi.org/10.1111/ecog.05534

To cite this report, use: Markus Konkol (2021): CODECHECK of *When is variable importance estimation in species distribution modelling affected by spatial correlation?* DOI: 10.5281/zenodo.5574909

Summary: The reproduction was (partially) successful. The authors provided the source code via a link to a shared folder. From the three relevant figures in the paper, two were reproducible (Figure 4 and 5) with only minor deviations from the corresponding figures reported in the paper. The deviations were caused by randomness. The authors demonstrated concern for the reproducibility of their work by improving the code and responding to requests throughout the review process.

Date of check: 18th October, 2021

CODECHECKER notes

The paper is already published and does not contain a link to the source code. However, the authors stated that the code will be made available via 4TU. They are also in contact with the editors of the paper in order to add the link to the paper. The authors reserved a DOI for that: 10.4121/16752223

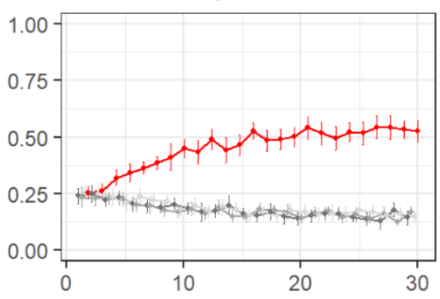
The authors provided me with a link to a shared folder containing the code. ***Recommendation:*** *Use a collaborative versioning software for sharing code (e.g., git) to allow forking and pull requests.* A first execution attempt resulted in an error because of missing libraries. All libraries needed were available on the package system *CRAN*. The second error was caused by a missing functionality. The missing function was defined in a separate script (”*all_functions.R*”). ***Recommendation:*** *Use the function source(“path_to_file”) to load scripts in the main file.* Another issues was caused by the file directory used in the function *setwd()*. I changed the file path from absolute to relative paths. ***Recommendation:*** *Use relative instead of absolute file paths to make directories independent of the system*. From the script and the documentation, it was not clear to me, how to set the parameters to reproduce the figures. Through a request to the authors, I got all relevant information. ***Recommendation:*** *Document the parameter settings needed to reproduce all figures, too*. *If time allows, create an interactive figure, e.g., using Shiny.* Apart from these minor issues, the script could be executed smoothly. ***Recommendation:*** *Use R Markdown to generate an output document including all figures.* After I changed the parameter settings, I was not able to execute the entire code again due to the following error: Error in rescale(gawd@data@values, mean = 100, df = F) : unused arguments (mean = 100, df = F). A restart of RStudio solved this issue but it might require revision.

Below are all reproduced and original figures. As indicated before, the figures are not exactly the same because of some randomness in the code. However, the curves indicate a similar tendency.

Reproduced figures

Figure 4a)

Reproduced:



Original:

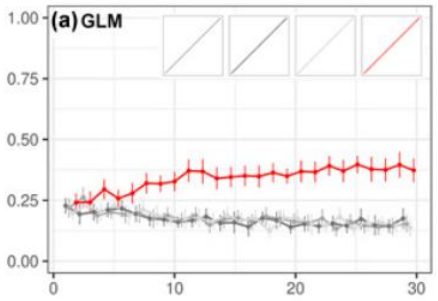
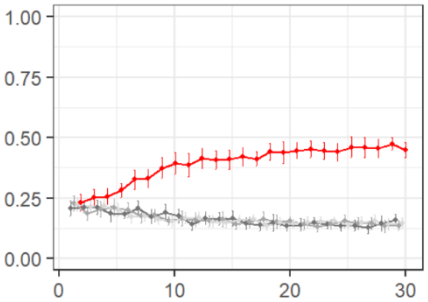


Figure 4b)

Reproduced:



Original:

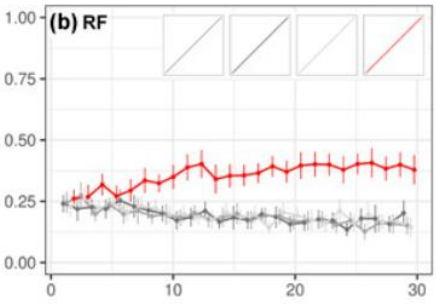
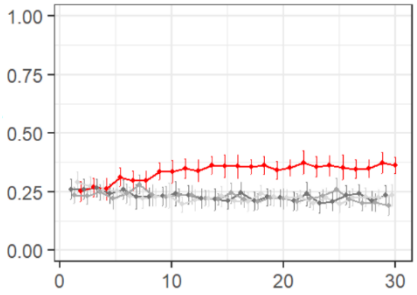


Figure 4c)

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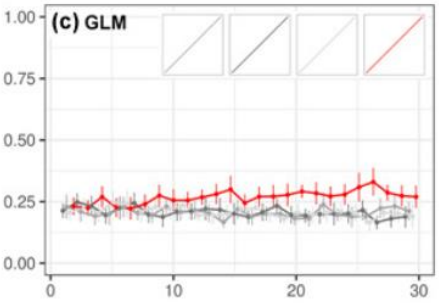
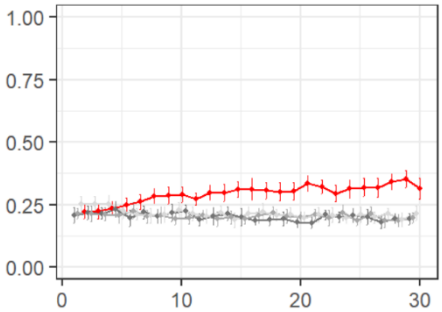


Figure 4d)

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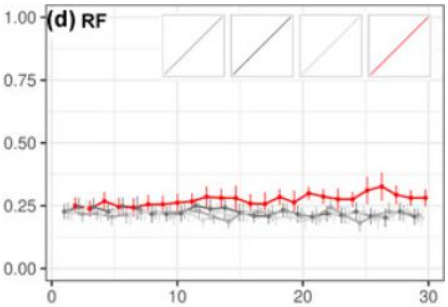
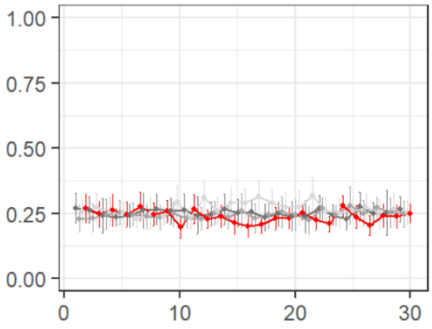


Figure 4e)

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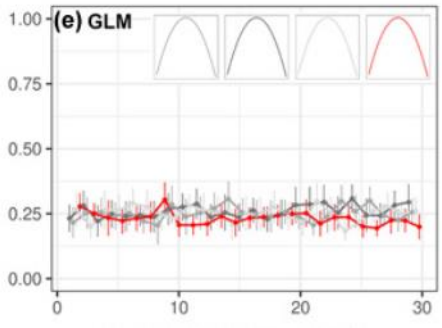
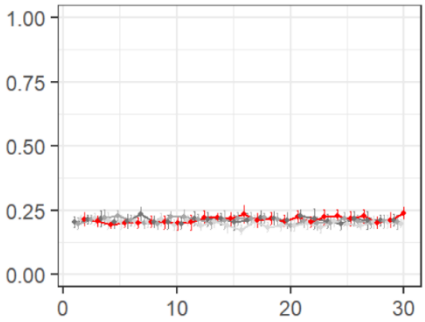


Figure 4f)

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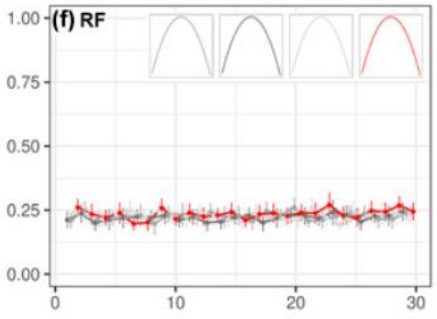
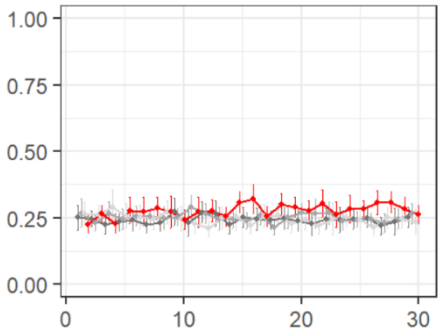


Figure 4g)

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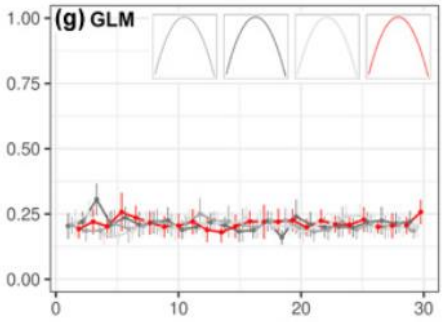
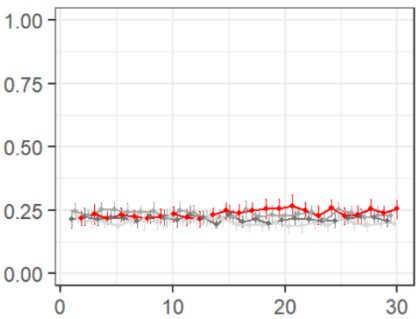


Figure 4h)

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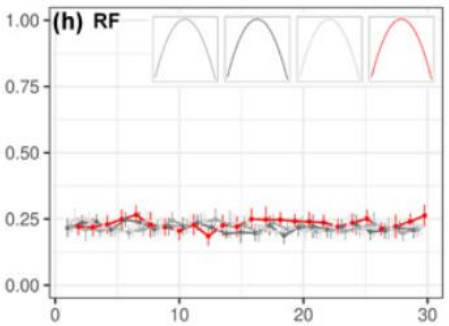
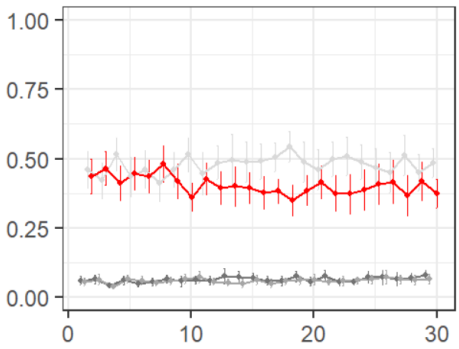


Figure 5a)

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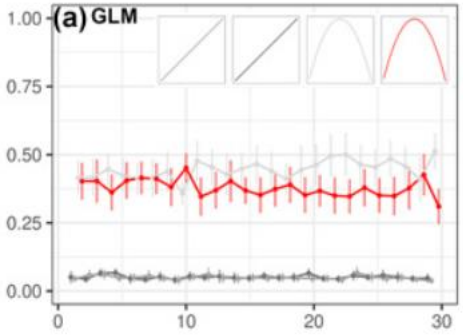
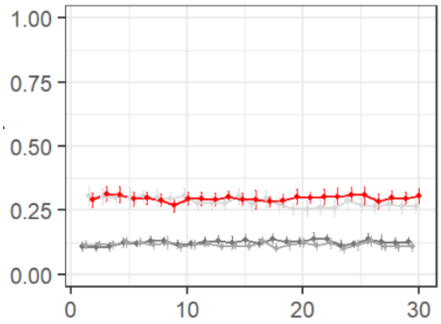


Figure 5b)

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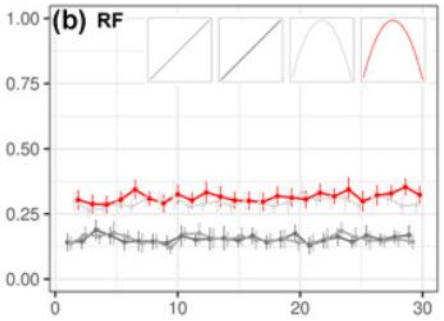
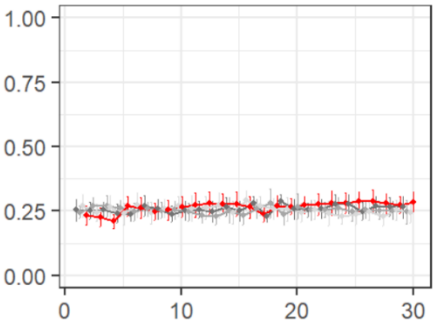


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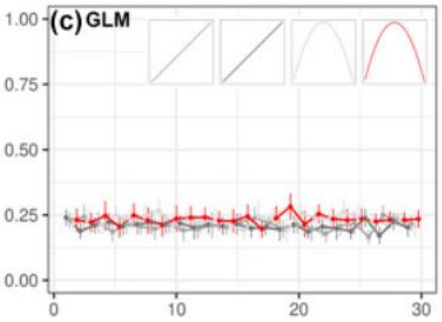
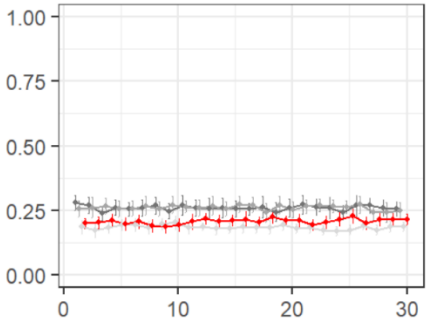


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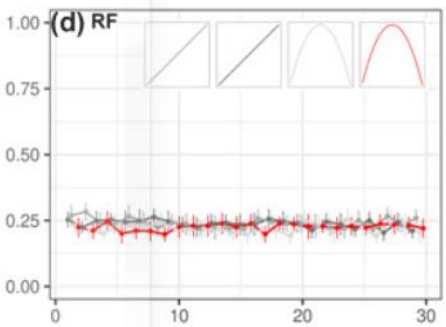
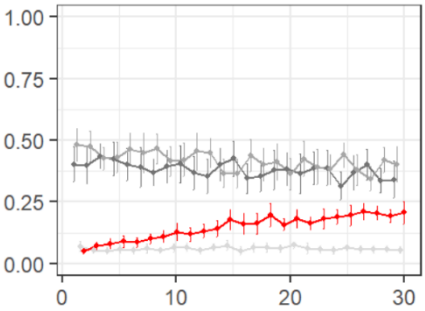


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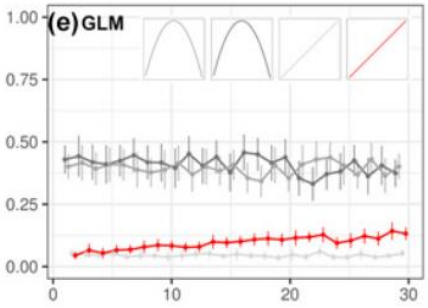
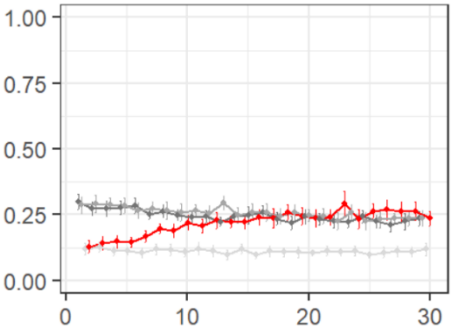


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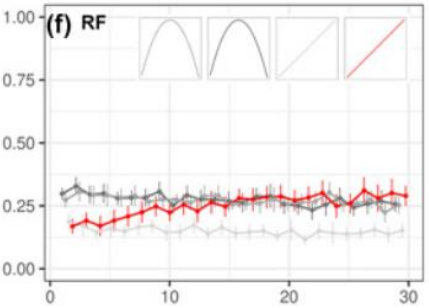
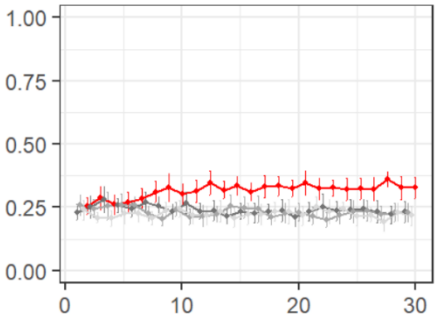


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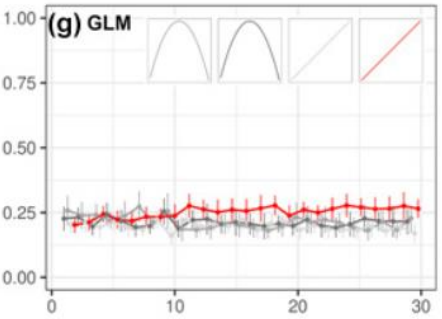
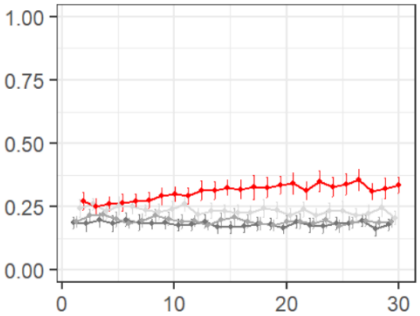
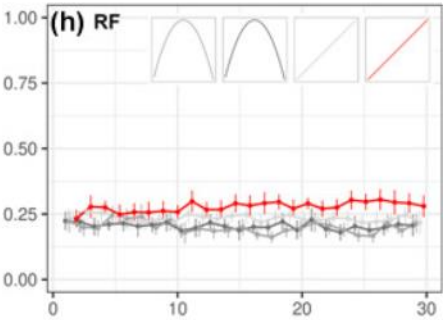


Figure 5h)

Reproduced:



Original:



Used computational environment

Windows 10 Enterprise, Intel Core i7-8665U CPU 1.90GHz 2.11GHz, 32GB RAM, 64-bit Operating System. R version 4.0.3, RStudio Version 1.4.1103.